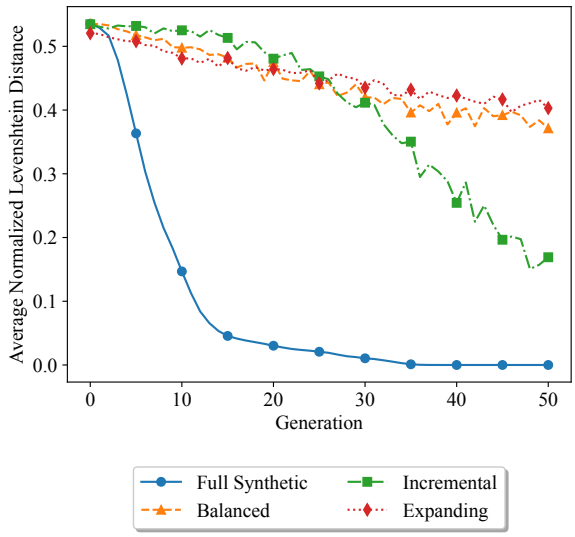

Figure 4: Average pairwise normalized Levenshtein distance over generations of a self-consuming training loop for different data cycles. Line markers added every 5 generations for better display.

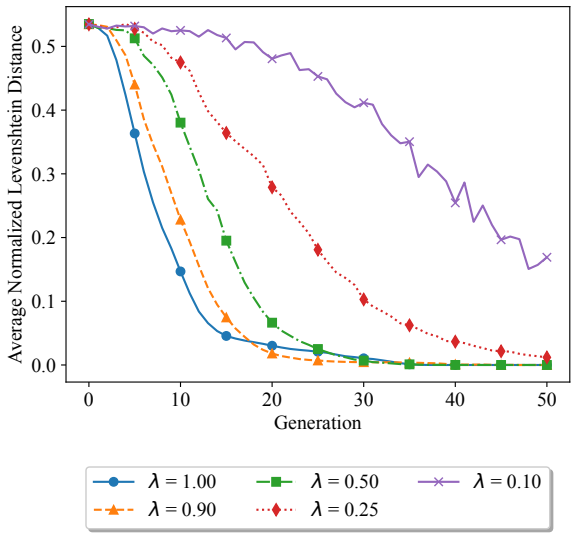

Figure 5: Average pairwise normalized Levenshtein distance over the course of a self-consuming training loop for the incremental data cycle with different portions λ of new sampled data. Line markers added every 5 generations for better display.

For the full synthetic data cycle we see a steep decline in diversity with a collapse into a single point by generation $t = 39$. The incremental data cycle is initially stable, but decreases in diversity from the 10th generation onwards, with a decrease of 68% in diversity by the end of the 50 generations. The balanced and expanding data cycle also decrease in diversity, however, this decrease is way slower, with a 30% decrease in diversity for the balanced data cycle and a 22% decrease for the expanding data cycle. While not all data cycles fully collapse in diversity by generation 50, ultimately, we expect all of them to eventually reach zero diversity if the self-consuming training loop is run for enough generations.

To get further insight into the decline in diversity, we also inspect the number of unique expressions sampled per generation for the full synthetic data cycle. We find that the number of unique expressions is stable at first while the diversity already declines and then rapidly decreases within 5 generations to very few unique expressions (see Appendix B).

To better understand the influence of the amount of generated data in each generation on the diversity, we also study the incremental data cycle in more detail with different portions λ of new sampled data. Figure 5 displays the average pairwise normalized Levenshtein distance over generations in a self-consuming training loop for the incremental data cycle with different values of λ. We observe, that the loss in diversity is stronger for a higher share of generated data added in each generation of the self-consuming training loop. Even with as little as $\lambda = 0.25$, we reach the complete loss of diversity within 50 generations. Only for our experimental run with $\lambda = 0.10$ the diversity does not drop to zero, but to a value around 0.17 after 50 generations. However, we see that the speed at which diversity decreases does not scale linear with the amount of generated data. Instead we observe a more exponential increase in speed where larger amounts of generated data lead to an even quicker decay in diversity.

Lastly, we study the expanding data cycle in more detail with regard to the proportion between generated data and fresh data added to the training data at each generation. Figure 6 displays the average pairwise normalized Levenshtein distance over generations in a self-consuming training loop for the expanding data cycle with different portions λ of generated data and $(1 - \lambda)$ fresh real data. We observe, that while each configuration declines in diversity, the rate of this decline depends on the proportion between generated and fresh data. The more fresh data is added at each generation, the slower diversity decreases. With no fresh data or only 1% of fresh data the diversity decreases by approximately 36%. If the portion of fresh data

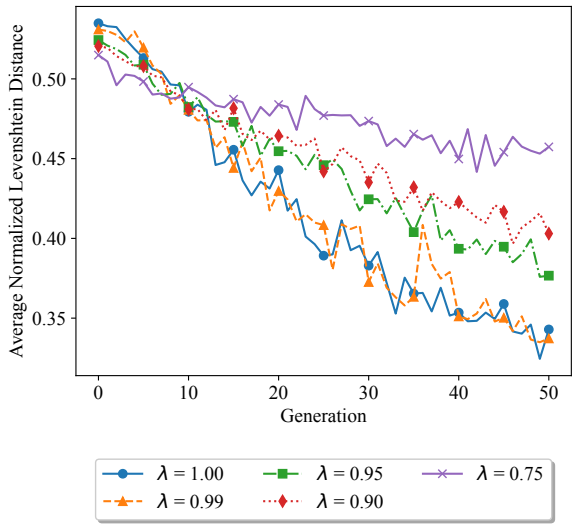

Figure 6: Average pairwise normalized Levenshtein distance over the course of a self-consuming training loop for the expanding data cycle for different portions λ of generated data and $(1-\lambda)$ of fresh data. Line markers added every 5 generations for better display.

is 25% we only observe a decrease in diversity of approximately 11%. This indicates that fresh data cannot stop the self-consuming training loop but enough fresh data can slow down the rate of the decline in diversity.

5 Discussion

In our experiments we found that iteratively retraining LLMs from scratch might help with correctness of model output. This is sometimes already done in practice where the output of LLMs is used to improve their performance (Huang et al., 2022; Wang et al., 2022) or generate new training data (Yu et al., 2023). However, this trend is concerning as we also found that this self-consuming training loop eventually leads to a drastic loss of diversity of a model's output. Our results confirm the findings on the self-consuming training loop in the field of image generation (Martínez et al., 2023; Alemohammad et al., 2023; Shumailov et al., 2023; Bertrand et al., 2023), further highlighting the importance of this issue for generative models and LLMs in particular. These LLMs have an even greater impact on society than image generation models, leading to more LLM-generated data being produced and potentially having a greater impact on society as a whole, as many people already rely on good model performance.

Consequently, as we expect that a self-consuming training loop will harm model performance in the future, researchers and practitioners should carefully choose their data when training their models and test those models sufficiently for diversity. Additionally, the machine learning community at large needs to find ways to deal with this problem as generated text data is already present in new internet scraped datasets. While our results indicate that fresh real data can slow down the self-consuming training loop, other studies suggest that this is not an option as we will run out of real data eventually and have to rely on generated data for future models (Villalobos et al., 2022). Additionally, LLM-generated data has already found its way onto the internet, even infecting academic publications (Geng and Trotta, 2024; Liang et al., 2024), and LLM-generated data may soon overtake real data on the internet.

Furthermore, this LLM-generated data is hard to differentiate from real data (Kreps et al., 2022; Sadasivan et al., 2023; Huschens et al., 2023), making it difficult to curate datasets scraped from the web. This makes the self-consuming training loop inevitable in the future. Therefore, we advise to further study ways to maintain diversity of generative model outputs like quality diversity methods (Pugh et al., 2015; Fontaine and Nikolaidis, 2021).

6 Conclusion

In this paper we studied the behavior of LLMs trained in a self-consuming training loop and found that iteratively training LLMs from scratch with self generated data can initially help with correctness of model outputs. However, this comes at a price, as the diversity of generated data eventually degenerates and collapses into a single point. The rate of this decline in diversity depends on the data cycle creating the training data at each generation. Furthermore, we found that fresh data can slow down this process and preserve diversity of a model output for longer. This should encourage researches to carefully look at their datasets and models to mitigate such a harmful self-consuming training loop in the future.

In future work, we will study how fine-tuning can influence the self-consuming training loop for LLMs. Additionally, we plan to further investigate how the effects of a self-consuming training loop can be slowed down or completely negated.

7 Limitations

One major cause for limitations of our work are limited computational resources. First, we ran our experiments with a smaller GPT-style model than for example GPT-4. Even though we expect that larger models with billions of parameters still suffer from the same loss of diversity if trained in a self-consuming training loop, we note that the behavior may vary for those larger models. However, a study for models of those size is not feasible with current compute options nor is it responsible in terms of the expected CO_2 emissions (Strubell et al., 2019).

Second, another limitation imposed by our computational resources is that the main results of this work consist of only one run per experimental configuration. With more runs per experiment the results would be more robust, but conducting more runs for our experiments was simply not feasible with our available computing resources. As all our experiments point towards the same direction, however, we do not believe that this limitation is crucial. Nevertheless, to further mitigate this limitation we performed additional experiments for the full synthetic data cycle with fewer generations over multiple runs and with different initializations in Appendix C. These results show that the trend of a decline in diversity still holds true over multiple runs.

Another limitation is that our work focuses on repeatedly training foundation models from scratch in a self-consuming way. However, in practice those models are normally fine-tuned after the initial training. It is unclear from our work if this fine-tuning process can slow down or negate the effects of a self-consuming training loop when performed with fresh real data. However, we plan to investigate this in future work.

Lastly, one may argue that real text has a way larger vocabulary and more complex structures. Therefore, our results might be slightly different when performed on a real text corpus. However, measuring both quality and diversity of generated text is not trivial and can only achieved by using a proxy (Callison-Burch et al., 2006; Gehrmann et al., 2023). Consequently, we intentionally performed our study on a logic expression dataset to ensure a controlled experimental setting in which we can measure both quality and diversity of a trained model.

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